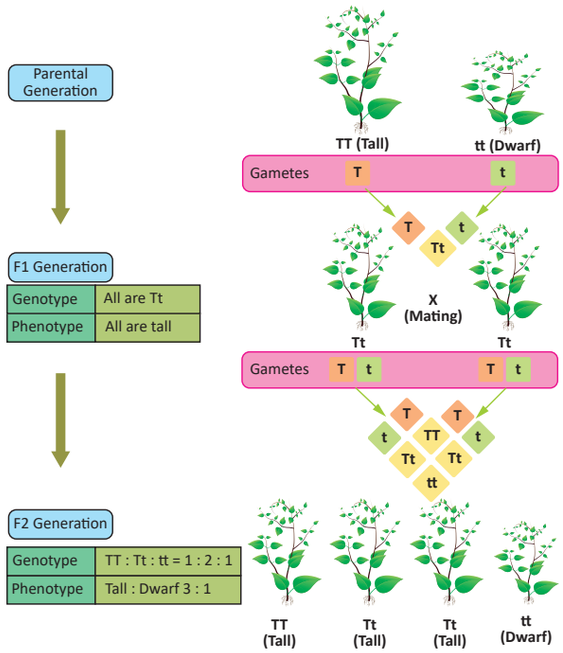


Figure 18.1 Monohybrid cross

in pairs and may be alike as in pure breeding tall plants (TT) and dwarf plants (tt). This is referred to as **homozygous**. If they are unlike (Tt) they are referred to as **heterozygous**.

1. Two factors making up a pair of contrasting characters are called **alleles.** Phenotypic expression of alleles are called **allelomorphs**. One member of each pair is contributed by one parent.
2. When two factors for alternative expression of a trait are brought together by fertilization. The **character which expresses itself** is called **dominant (Tallness)** condition and that which is **masked** is called **recessive condition (Dwarfness)**.
3. The factors are always pure and when gametes are formed, the unit factors segregate so that each gamete gets one of the two alternative factors. It means that factors for tallness(T) and dwarfness(t) are separate entities and in a gamete either T or t is present. When F_1 hybrids are self crossed the two entities separate and then unite independently, forming tall and dwarf plants.

Info bits

Punnett square is a checker board form devised by a British geneticist R.C.Punnett for study of genetics. It is a graphical representation to calculate the probability of all possible genotypes of offsprings in a genetic cross.

18.3 Dihybrid Cross- Inheritance Two Genes and Law of Independent Assortment

Dihybrid cross involves the **inheritance of two pairs of contrasting characteristics** (or contrasting traits) at the same time. The two pairs of contrasting characteristics chosen by Mendel were **shape and colour of seeds: round-yellow seeds and wrinkled-green seeds**.

Mendel crossed pea plants having round-yellow seeds with pea plants having wrinkled-green seeds. Mendel made the following observations:

1. Mendel first crossed pure breeding pea plants having round-yellow seeds with pure breeding pea plants having wrinkled-green seeds and found that only round-yellow seeds were produced in the first generation (F_1). No wrinkled-green seeds were obtained in the F_1 generation. From this it was concluded that **round shape and yellow colour of the seeds were dominant traits over the wrinkled shape and green color of the seeds.**
2. When the hybrids of F_1 generation pea plants having round-yellow seeds were cross-bred by self pollination, then four types of seeds having different combinations of shape and color were obtained in second generation or F_2 generation. They were

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